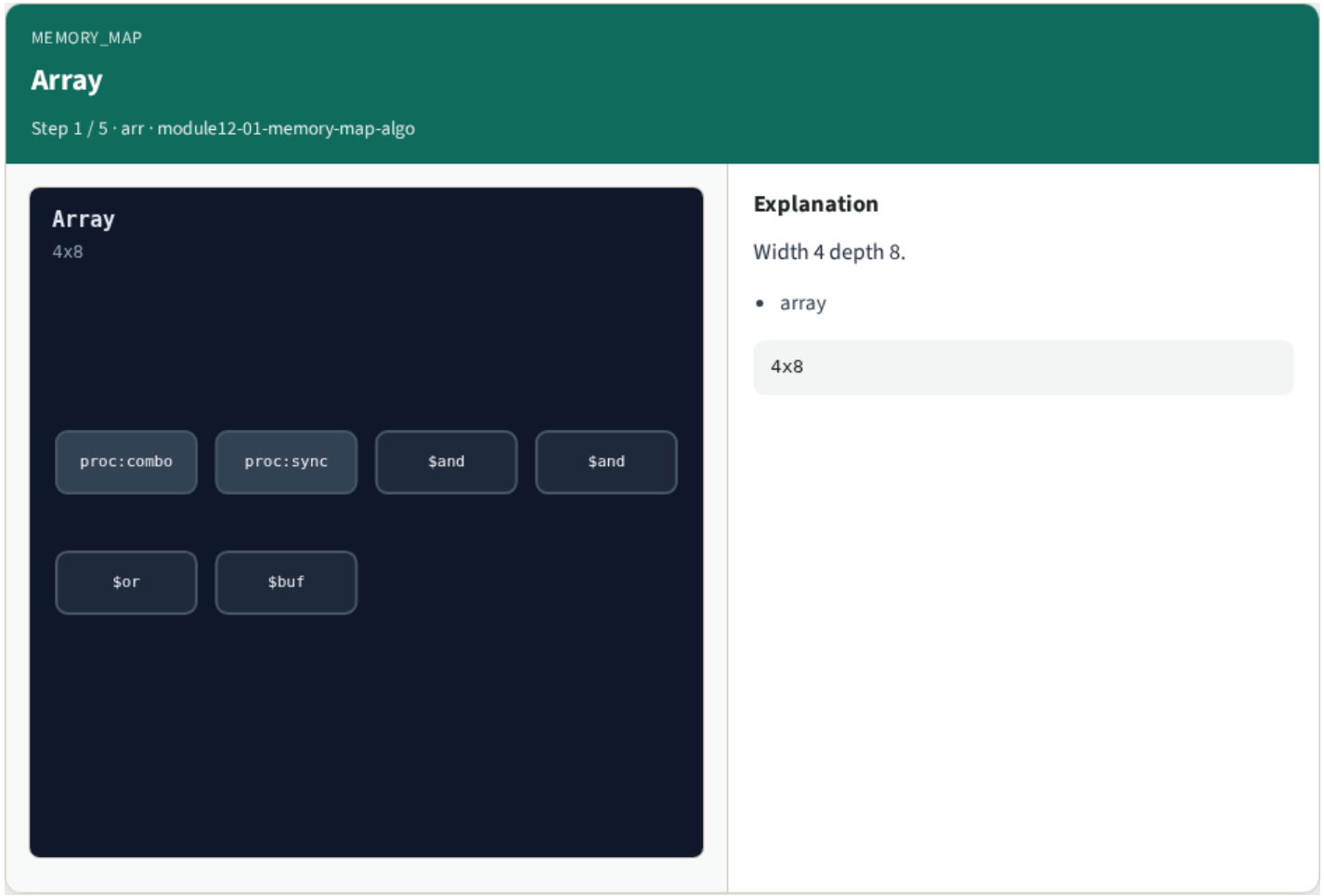
Why memory

Memory passes turn array descriptions into dollar-mem cells with size and width

The idea

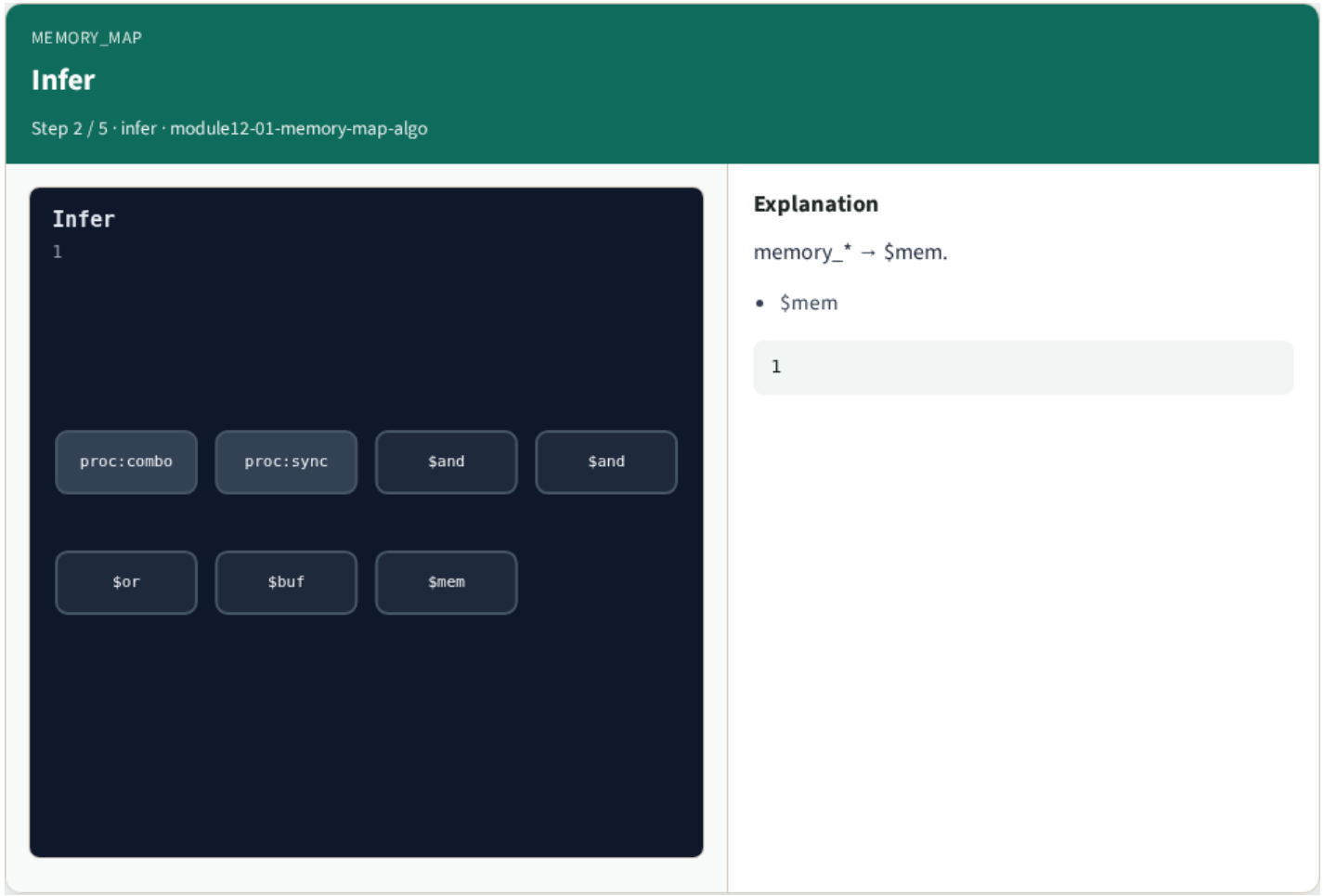
- GOLDEN: one dollar mem; depth=8; width=4

Array


MEMORY_MAP
Array
Step 1 / 5 · arr · module12-01-memory-map-algo
Array
4x8
proc:combo
proc:sync
$and
$and
$or
$buf
Explanation
Width 4 depth 8.
array
4x8

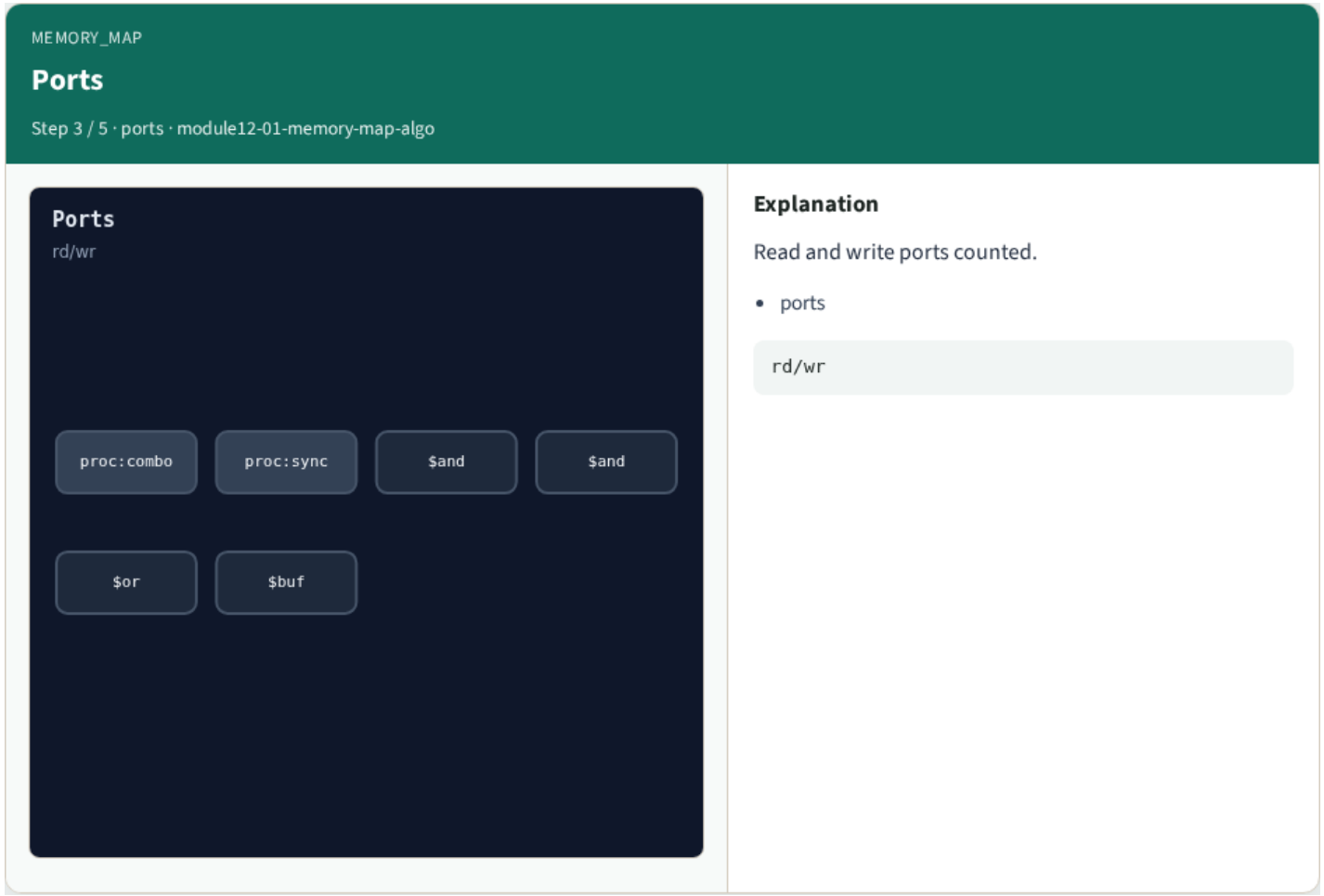
Array

Infer


MEMORY_MAP
Infer
Step 2 / 5 · infer · module12-01-memory-map-algo
Infer
1
proc:combo
proc:sync
$and
$and
$or
$buf
$mem
Explanation
memory_* → $mem.
$mem
1

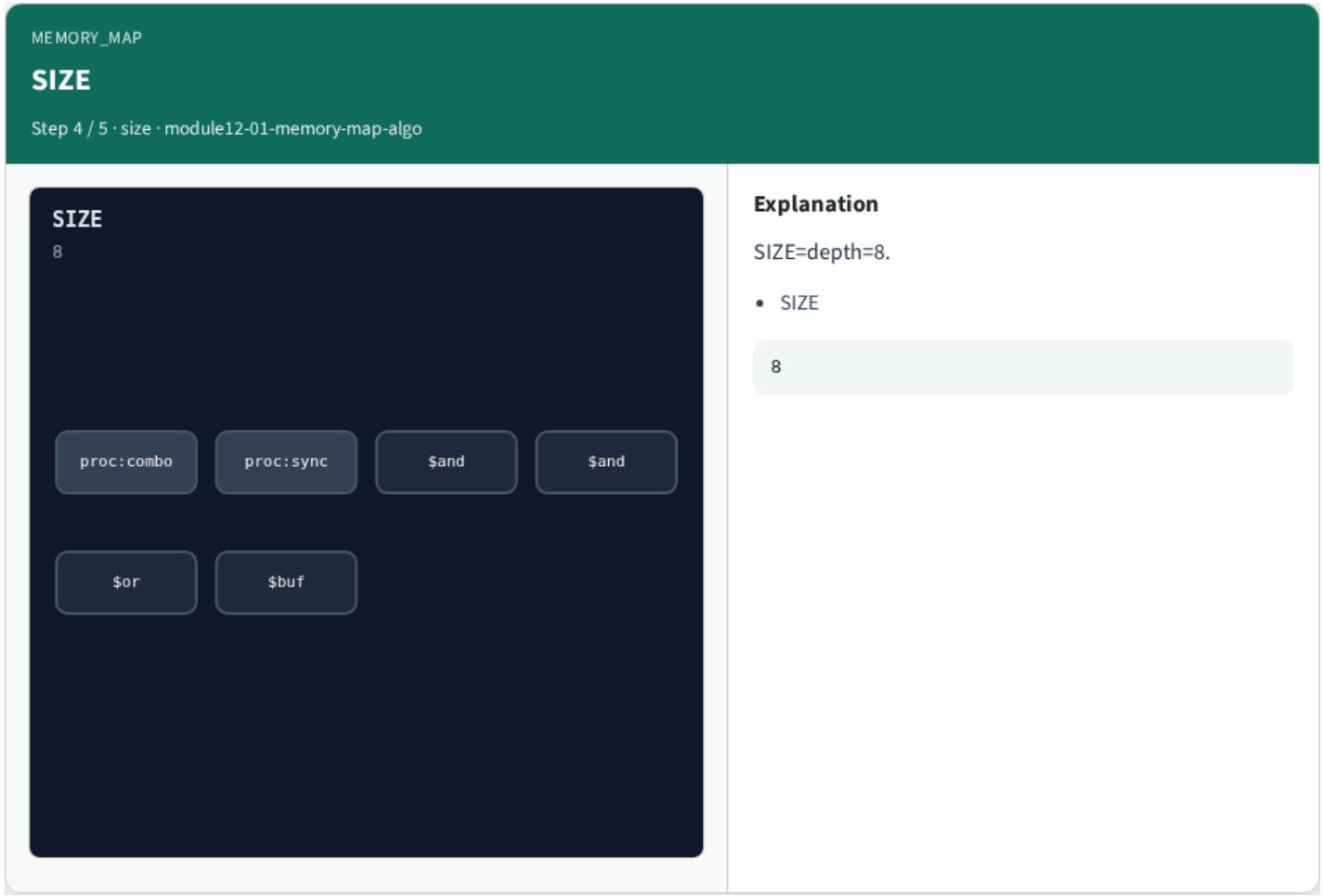
Infer

Ports


MEMORY_MAP
Ports
Step 3 / 5 · ports · module12-01-memory-map-algo
Ports
rd/wr
proc:combo
proc:sync
$and
$and
$or
$buf
Explanation
Read and write ports counted.
ports
rd/wr

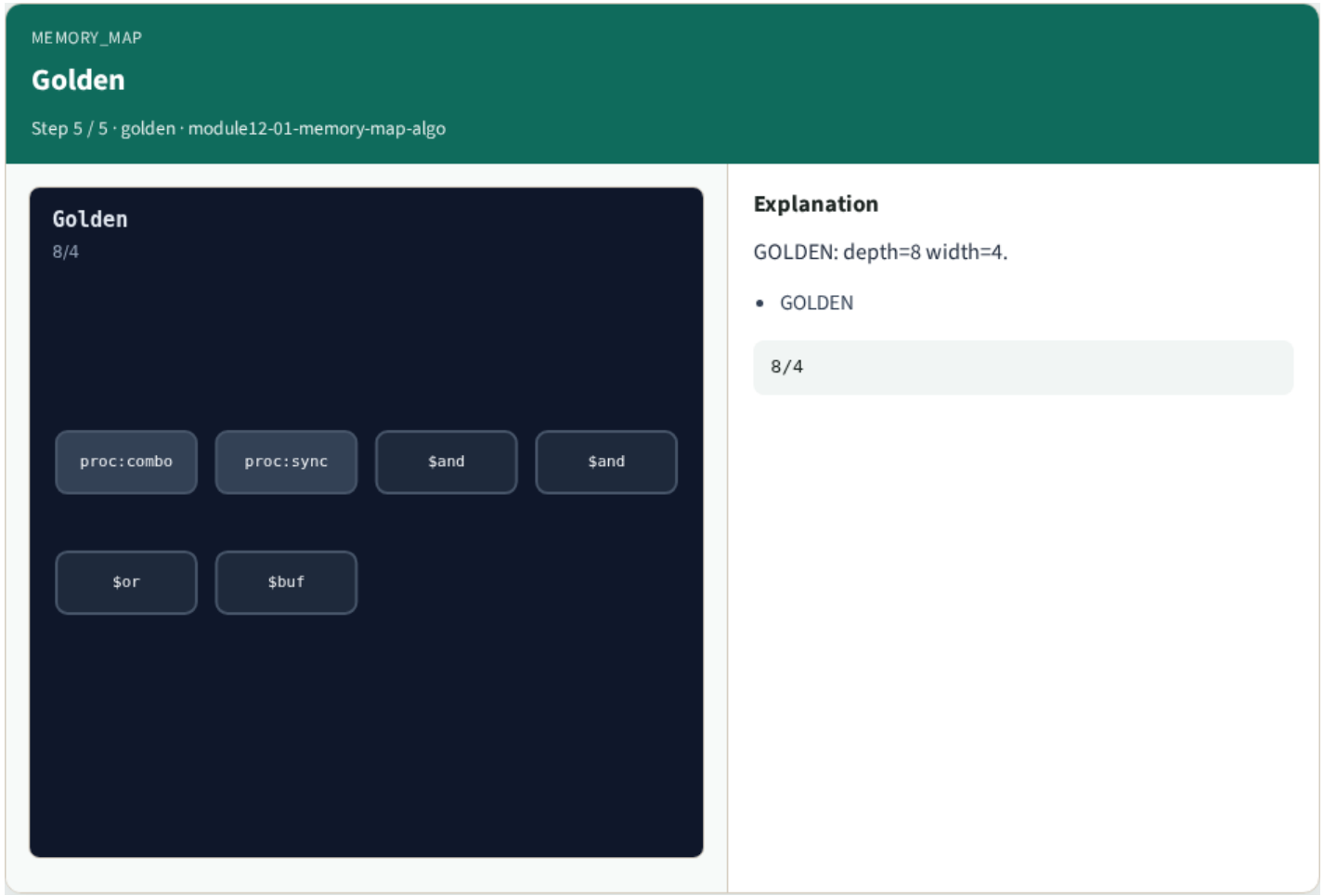
Ports

SIZE


MEMORY_MAP
SIZE
Step 4 / 5 · size · module12-01-memory-map-algo
SIZE
8
proc:combo
proc:sync
$and
$and
$or
$buf
Explanation
SIZE=depth=8.
SIZE
8

SIZE

Golden



Golden

Browser lab track

- Open **memory-map-algo**
- Build the mem cell and Check depth and width

Implement track

- Call algo_memory_map(); assert depth == 8 and width == 4

Pitfalls

- Swapping depth and width
- Ignoring read versus write ports

Try these

- From the common folder, run the algo solvers self-test, or call `algo_memory_map` directly

Try these — try these

```
cd courses/learn_yosys/common
python3 algo_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
read memory block from tiny_algo
emit dollar mem WIDTH=width SIZE=depth
ok if depth==8 and width==4
```

Next

- Complete the quiz for this part
- Next: FSM detect and encoding bits